

AMTI (NMTC) - 2012
GAUSS CONTEST - PRIMARY LEVEL

PART - A

1. a, b where $a > b$ are natural numbers each less than 10 such that $(a^2 - b^2)$ is a prime number. The number of such pairs (a, b) is
 (A) 5 (B) 6 (C) 7 (D) 8
2. The number of three digit numbers that are divisible by 2 but not divisible by 4 is
 (A) 200 (B) 225 (C) 250 (D) 450
3. A, B, C are single digits. In this multiplication B could be

$$\begin{array}{r} AB \times \\ 7 \\ \hline BCA \end{array}$$

- (A) 7 (B) 1 (C) 2 (D) 4
4. The base of a triangle is twice as long as a side of a square. Their areas are equal. Then the ratio of the altitude of the triangle to this base to the side of the square is
 (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) 2
5. Two sequences S_1 and S_2 are as under :

$$S_1 : \frac{2}{1}, \frac{4}{3}, \frac{6}{5}, \dots$$

$$S_2 : \frac{1}{2}, \frac{3}{4}, \frac{5}{6}, \dots$$

The n^{th} term of S_1 is $S_1 : \frac{2n}{2n-1}$ and the n^{th} term of S_2 is $\frac{2n-1}{2n}$. The value of the difference between the 2012th terms of S_1 and S_2 is

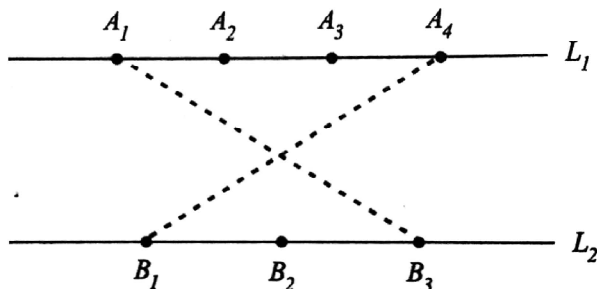
- (A) $\frac{4023}{2012 \times 2011}$ (B) $\frac{8047}{4024 \times 4023}$ (C) $\frac{4023}{4024 \times 4023}$ (D) $\frac{8047}{2012 \times 2011}$
6. The least number which when divided by 25, 40 and 60 leaves a remainder 7 in each case is
 (A) 607 (B) 1007 (C) 807 (D) 507
7. The integers greater than 1 are arranged in 5 columns as follows.

	Column (1)	Column (2)	Column (3)	Column (4)	Column (5)
Row 1 →		2	3	4	5
Row 2 ←	9	8	7	6	
Row 3 →		10	11	12	13
Row 4 ←	17	16	15	14	
⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮

In the odd numbered rows, the integers appear in the last 4 columns are increasing from left to right. In the even numbered rows, the integers appear in the first four columns are increasing from right to left. In which column will the number 2012 appears ?

- (A) fourth (B) second (C) first (D) fifth
8. Akash, Bharath, Christe, Daniel and Eashwar are friends. The interesting fact is that all of them were born in the same year, but on different days, different dates and different months. If Akash were born on February 19, then Daniel could have been born on
 (A) March 30 (B) August 20 (C) December 25 (D) April 16

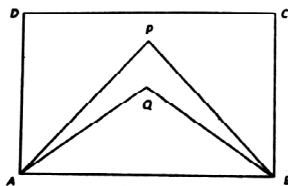
9. In the adjoining figure points A_1, A_2, A_3, A_4 are located on the line L_1 and B_1, B_2, B_3 are located on the line L_2 . Each one of the points on L_1 is connected to each one of the point of L_2 . (Example A_1 to B_3 and A_4 to B_1 as in the figure). The line segments are not extended. No line segment passes through the point of intersection of any two lines segments. The number of points of inter section of all these line segments is (Exclusive of A_1, A_2, A_3, A_4 and B_1, B_2, B_3).



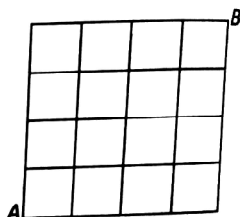
- (A) 15 (B) 16 (C) 17 (D) 18
10. A box contains 100 balls of different colours: 28 Red, 17 Blue, 21 Green, 10 white, 12 Yellow and 12 Black. The smallest number of balls drawn from the box so that at least 15 balls are of the same colour is
- (A) 73 (B) 77 (C) 81 (D) 85

PART - B

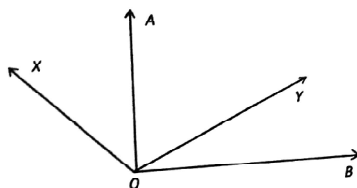
1. ABCD is a rectangle, AP, AQ divide $\angle DAB$ in to three equal parts and BP and BQ divide $\angle CBA$ into three equal parts. If $k(\angle APB) = (\angle AQB)$ then the value of k is _____.



2. Here is a sequence of composite numbers having only one prime factor, written in ascending order 4, 8, 9, 16, 25, 27, 32, The 15th number of this sequence is _____.
3. An insect crawls from A to B along a square lamina which is divided by lines as shown into 16 equal squares. The insect always travels diagonally from one corner of a square to the other corner. While going it never visits the same corner of any square. If one diagonal of a smallest square is taken as 1 unit, the maximum length of the path travelled by the insect is _____.

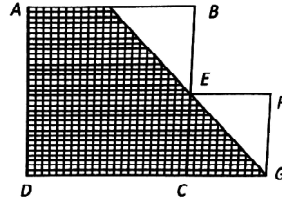


4. A says : "I am a 6-digit number and all my middle digits are made of zeros." B says to A : "I am your successor. My digit in the tens place is the same as your starting digit." The value of the whole number A is _____.
5. In the figure $\angle XOY = \angle AOB = 90^\circ$. The measure of $\angle XOY = 126^\circ$. The measure of $\angle AOY$ is _____.



6. 6 men can do a work in 1 year and 2 months. Then 3 men can do the work in _____ months.

7. The first term of a sequence of fraction is $\frac{3}{1}$ and the n^{th} term t_n of the sequence is equal to $\frac{\text{sum of the numerator and denominator of } t_{n-1}}{\text{Difference of the numerator and denominator of } t_{n-1}}$. (Ex. : If $t_1 = \frac{a}{b}$ and $t_2 = \frac{a+b}{a-b}$.)
The sum of this sequence to 2012 terms is _____.
8. In the figure ABCD and CEFG are squares of sides 6 cm and 2 cm respectively. The area of the shaded portion (in cm^2) is _____.



9. Master Ramanujan of Sixth standard was drawing squares of sides 1 cm, 2 cm, 3 cm and so on. After doing this for sometime he added the areas of the squares he made. He got the sum of the areas as 1015 cm^2 . The number of squares Ramanujan had drawn is _____.
10. The tens digit of a four digit number is an even prime. The number is divisible by 5. The other digits are all prime numbers and all the digits are distinct. The sum of all such four digit numbers is _____.
